

图 4.1-2

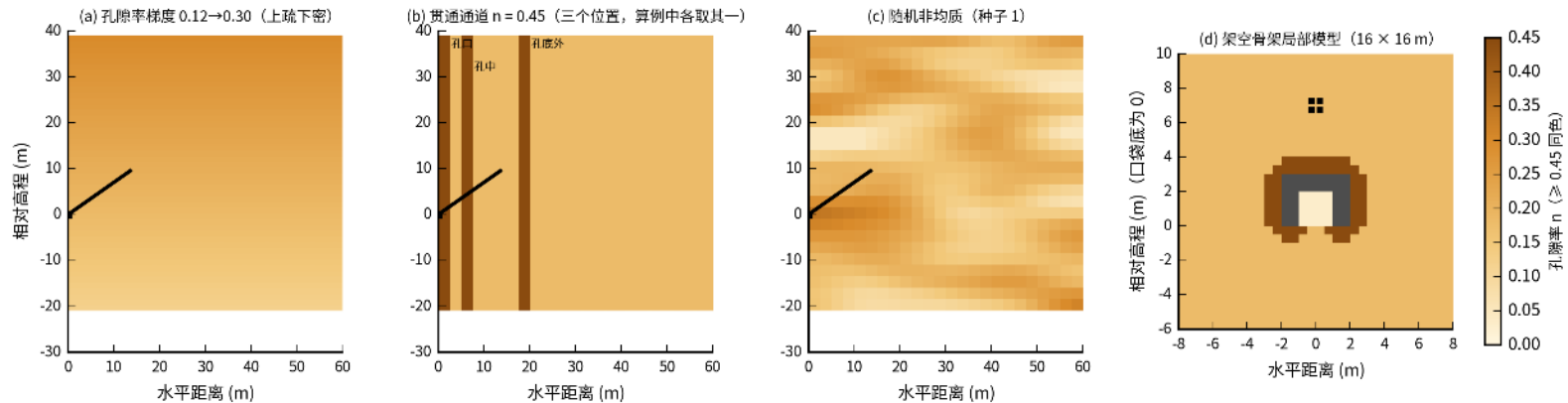


图 4.3-1

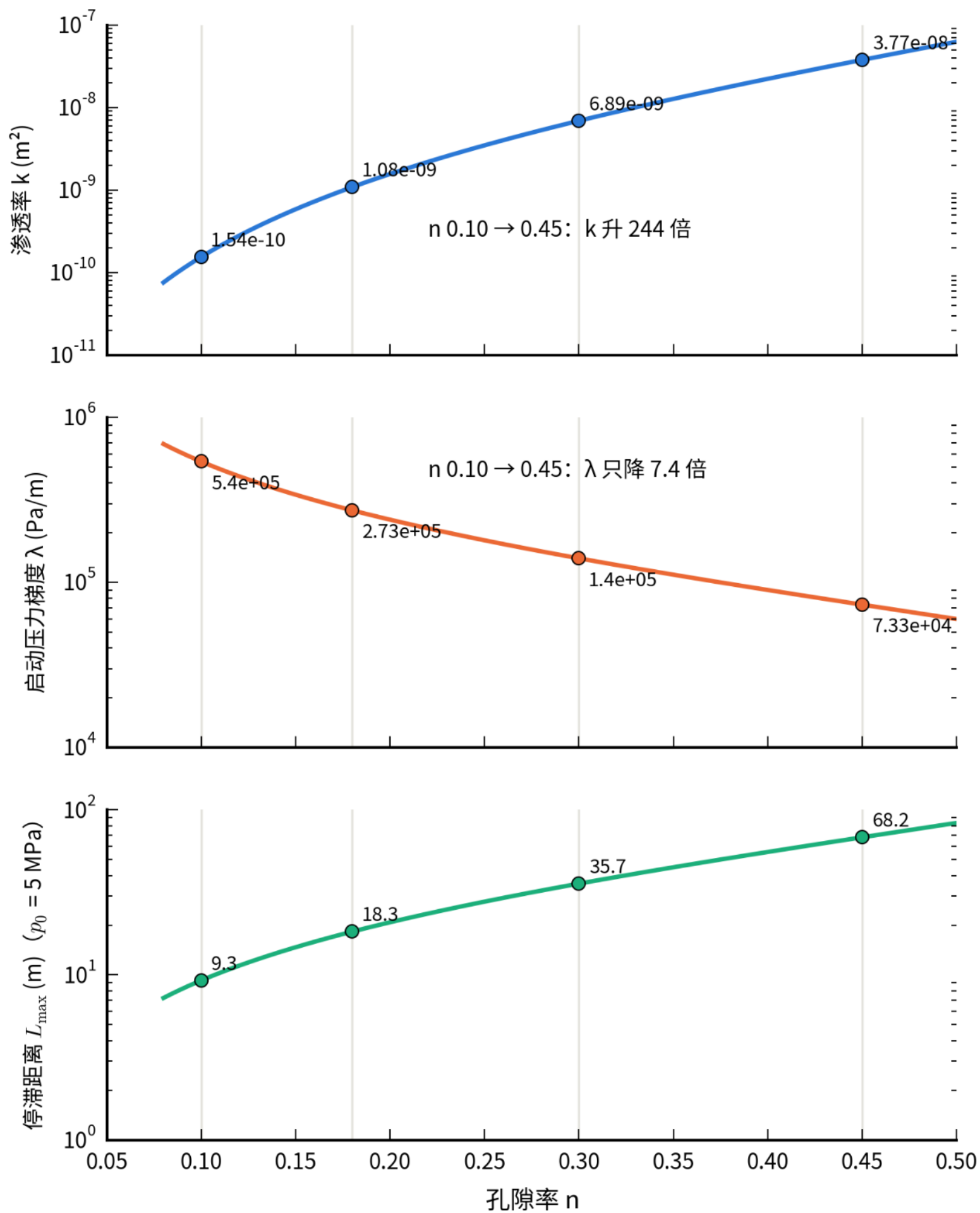


图 4.4-2

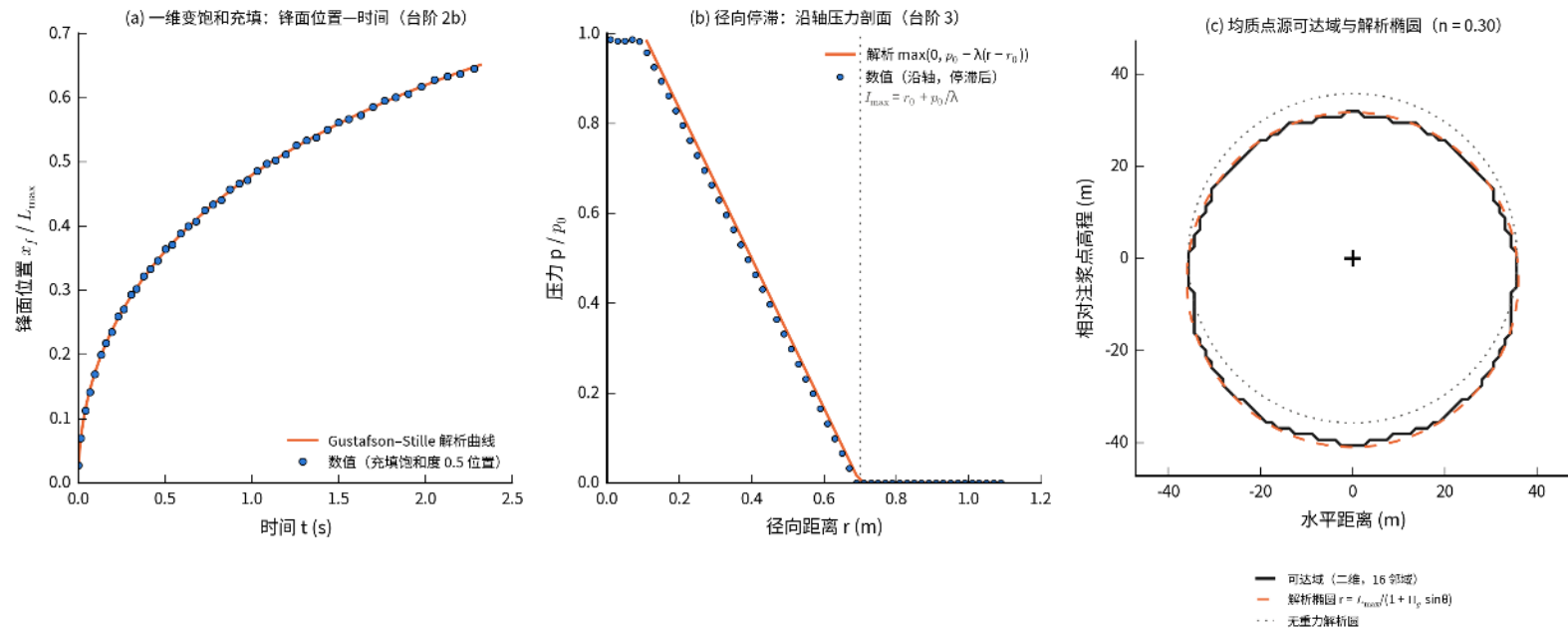
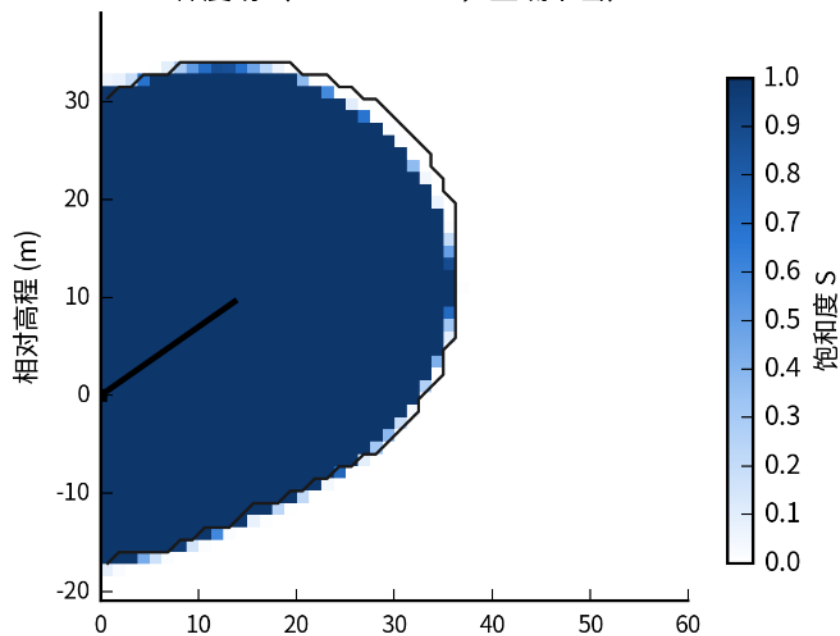
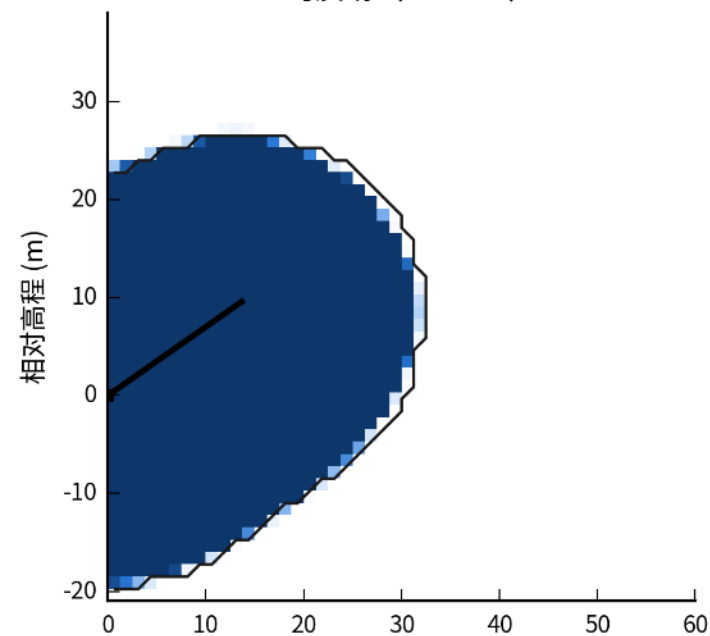


图 4.5-1

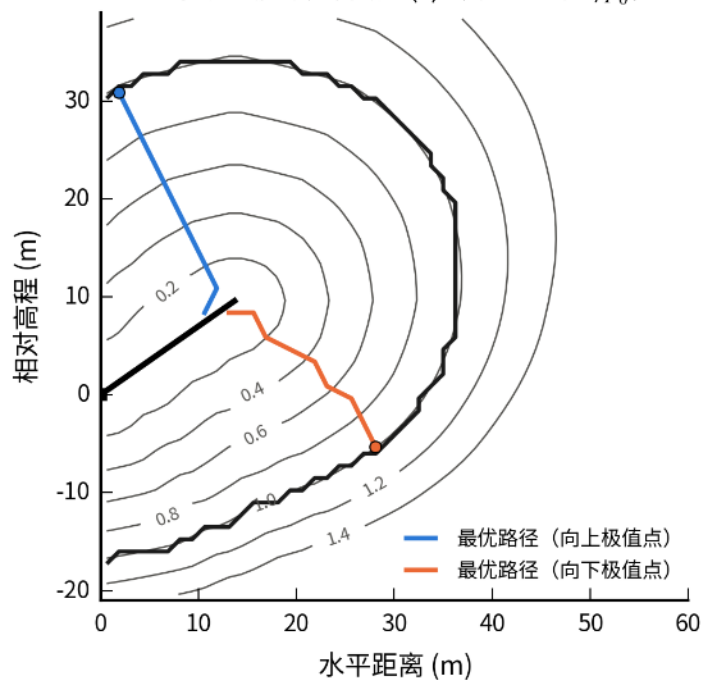
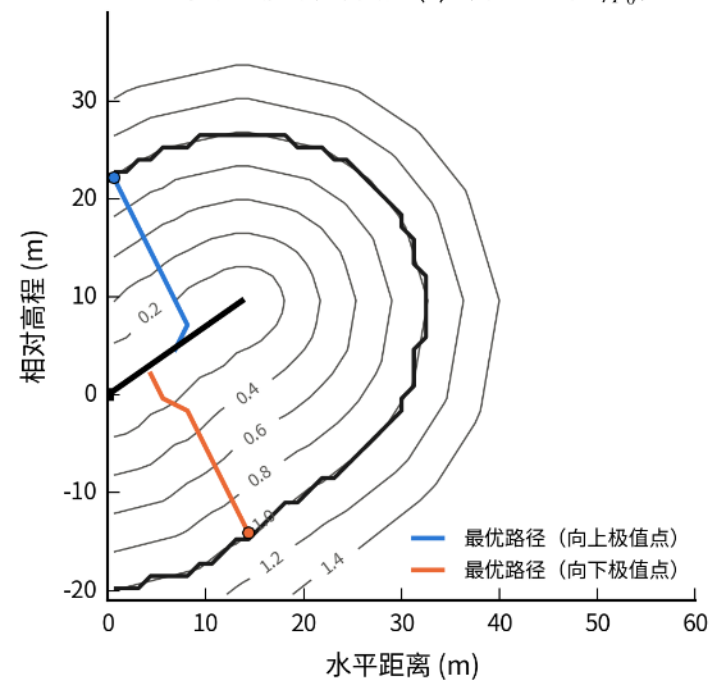
均质场 ($n = 0.18$)

梯度场 ($n\ 0.12 \rightarrow 0.30$, 上疏下密)



水平距离 (m)
最小累积启动压力场 $T(x)$ (等值线为 T/p_0)

水平距离 (m)
最小累积启动压力场 $T(x)$ (等值线为 T/p_0)



实线: 可达域; 色斑: 充填域 (停注时); 钻孔孔口位于左边界 (坡面)

图 4.5-2

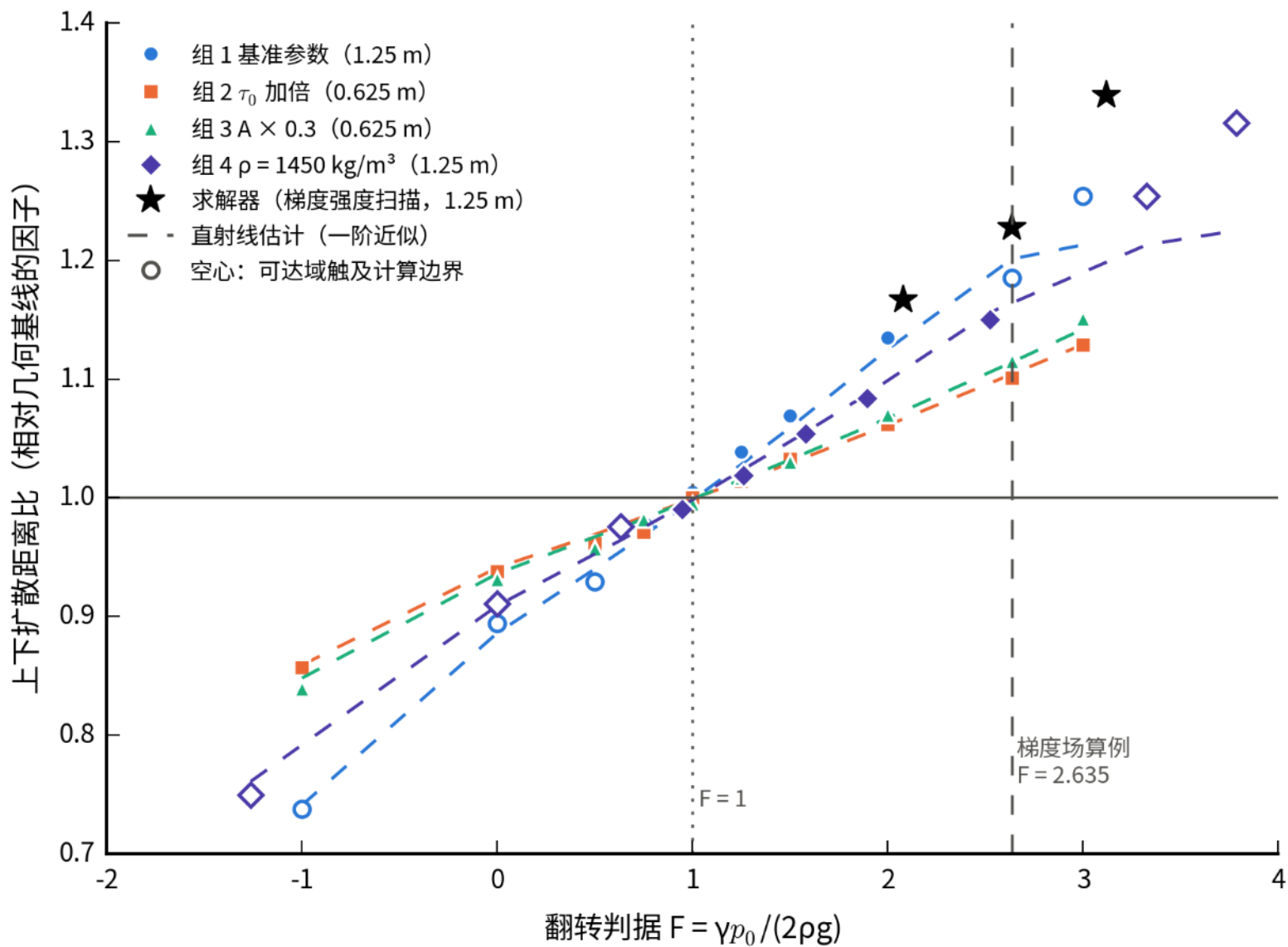


图 4.5-3

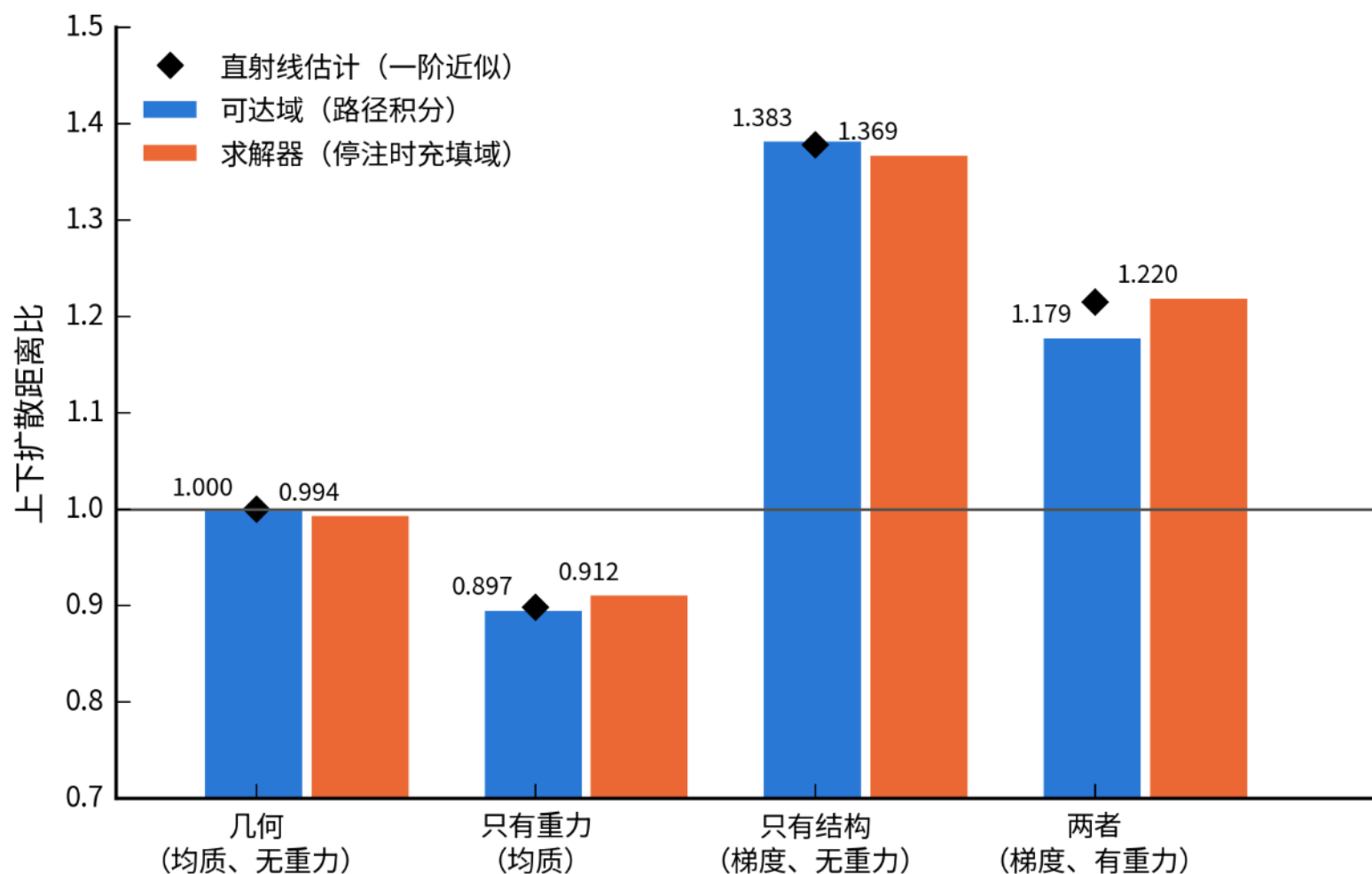


图 4.5-4

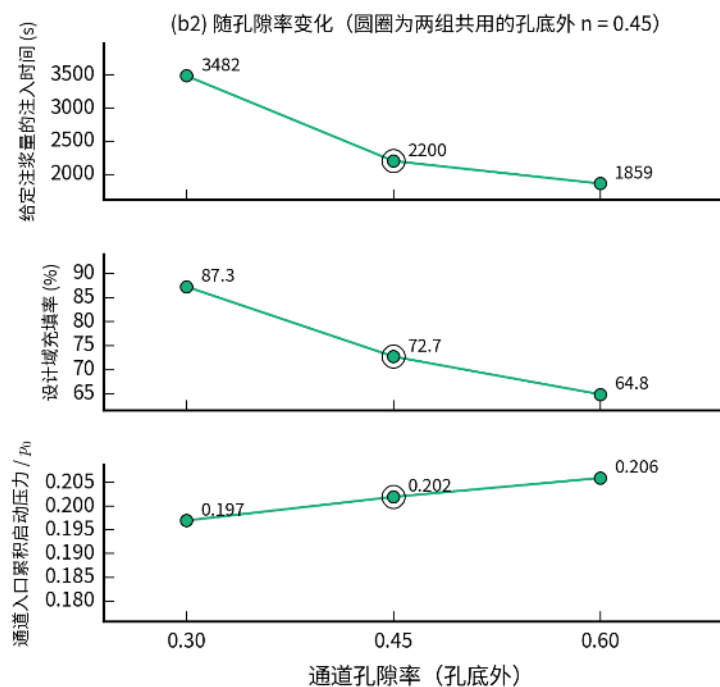
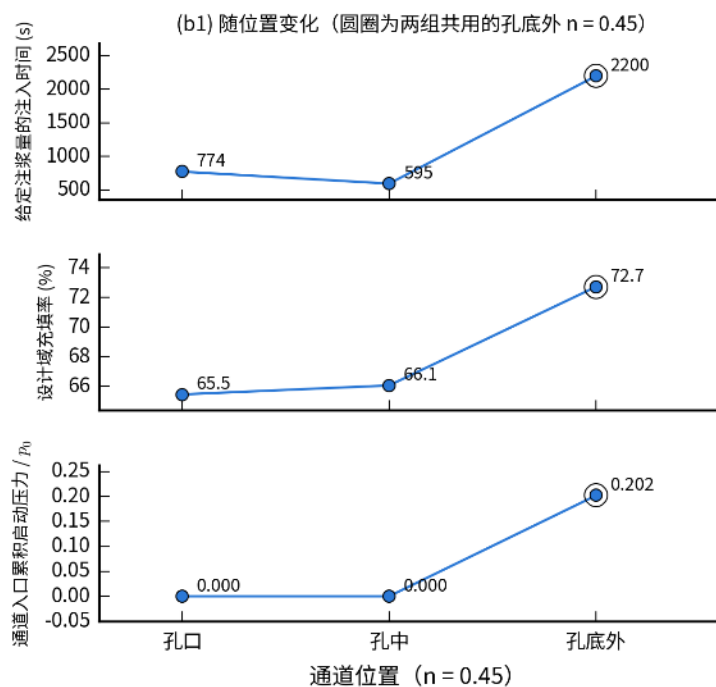
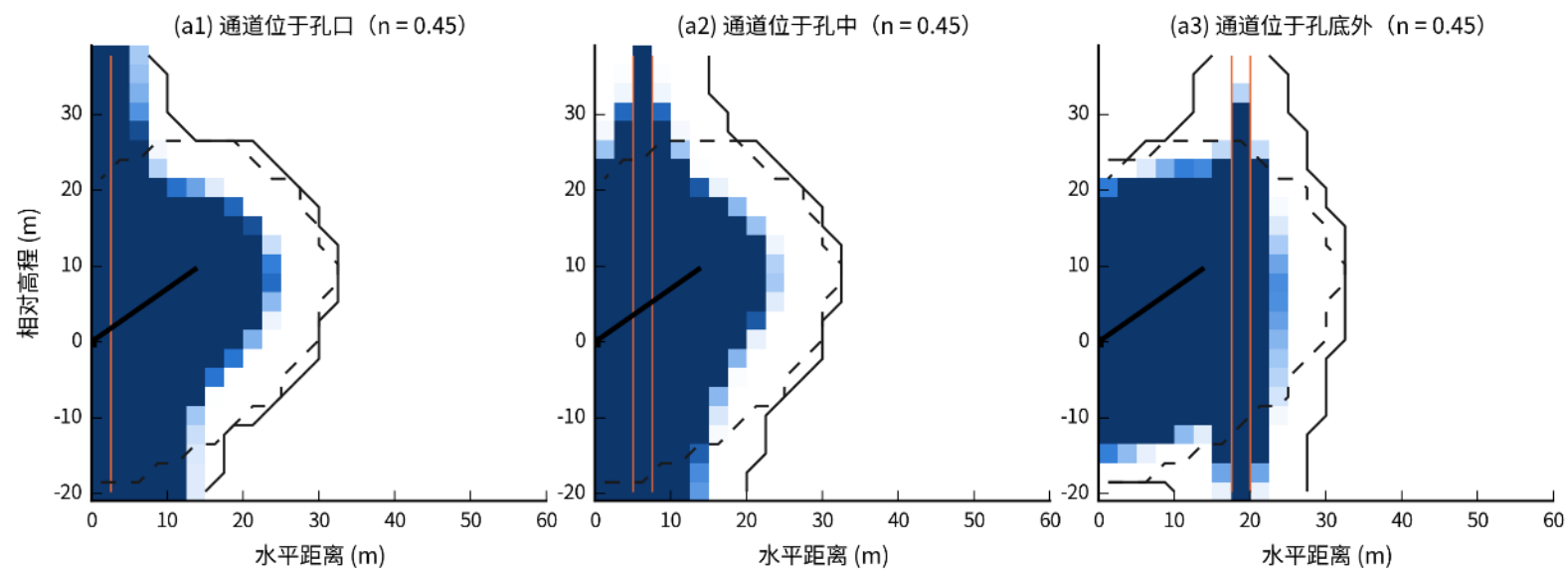


图 4.5-5

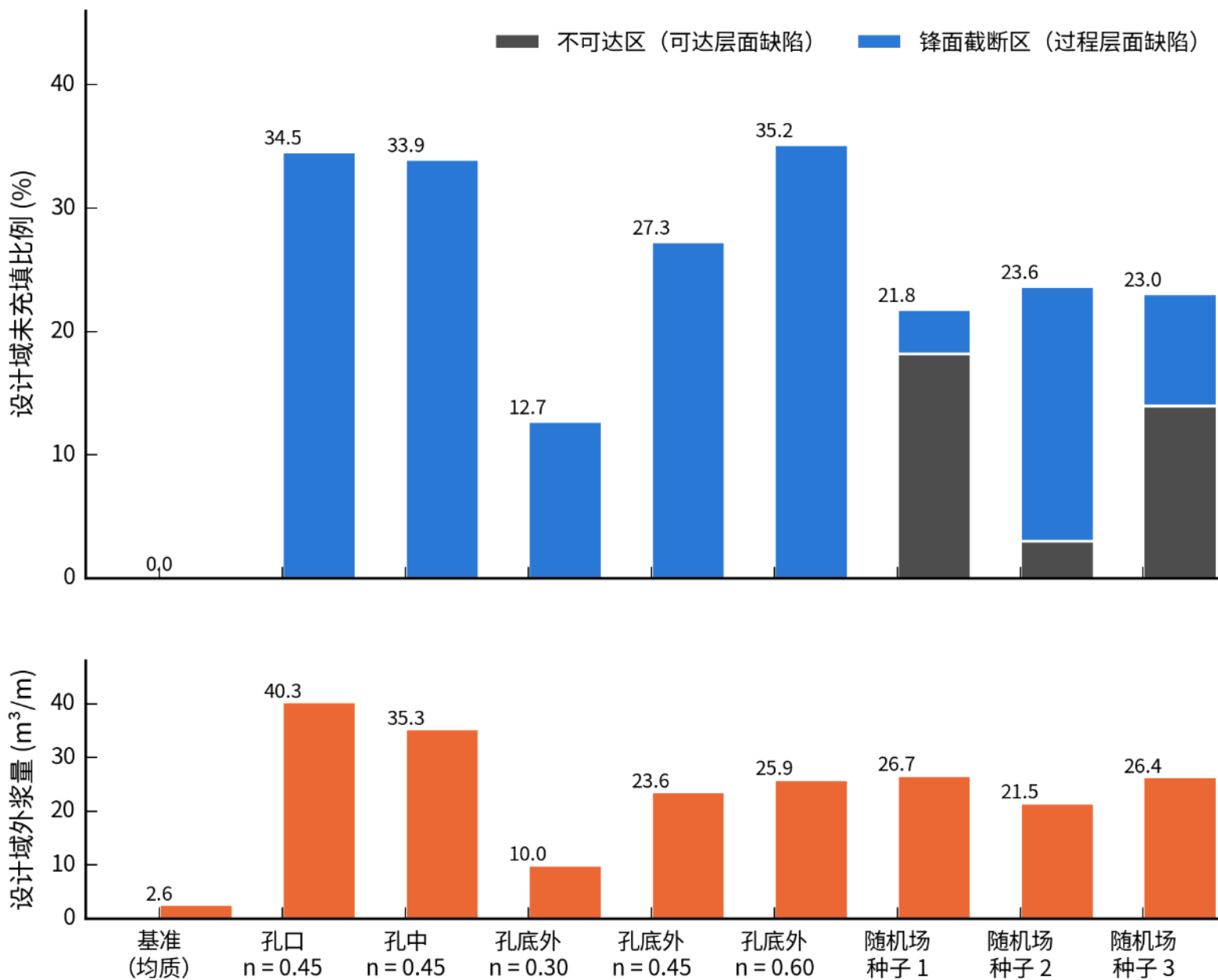
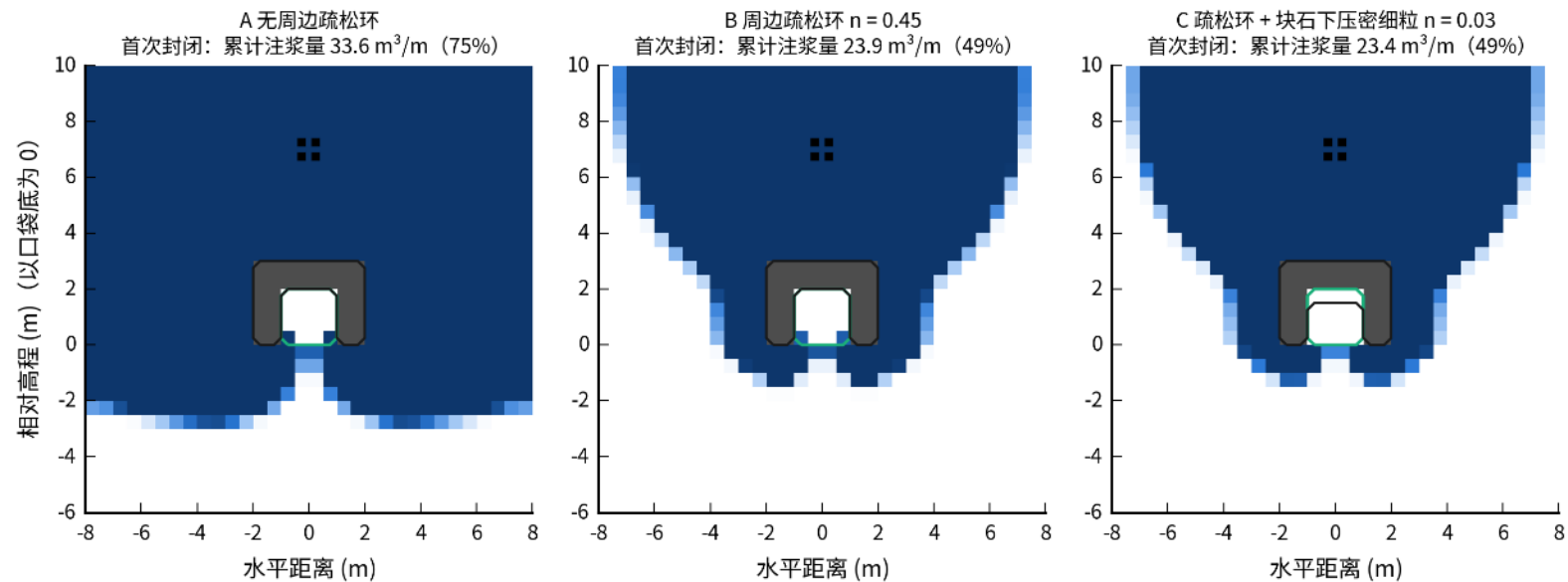


图 4.5-6



色斑: 充填域 (首次封闭时); 深灰: 块石; 绿线: 块石下口袋; 实线: 可达域; 黑方块: 注浆源

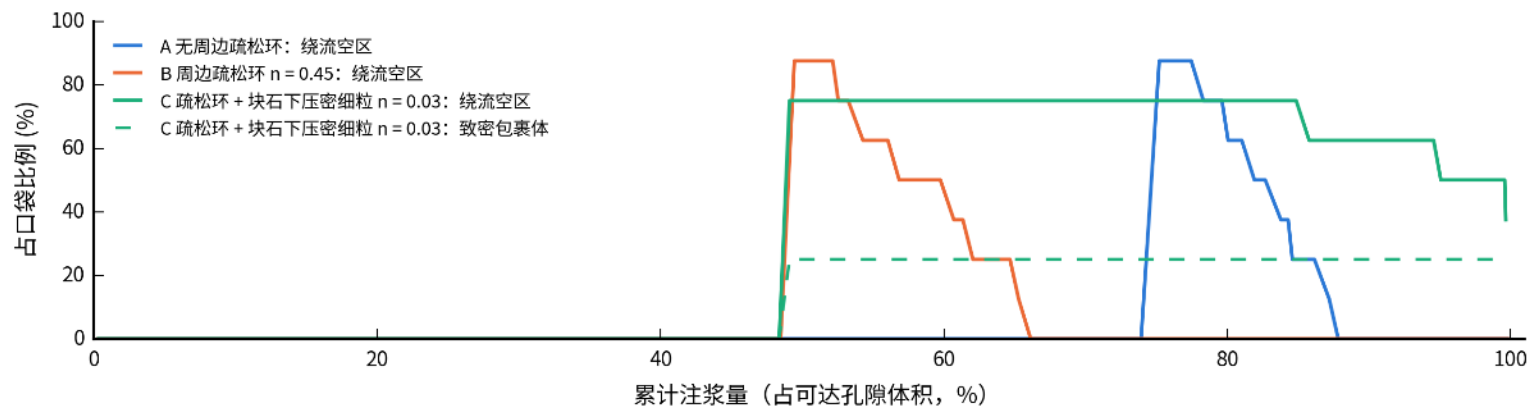


图 4.6-2

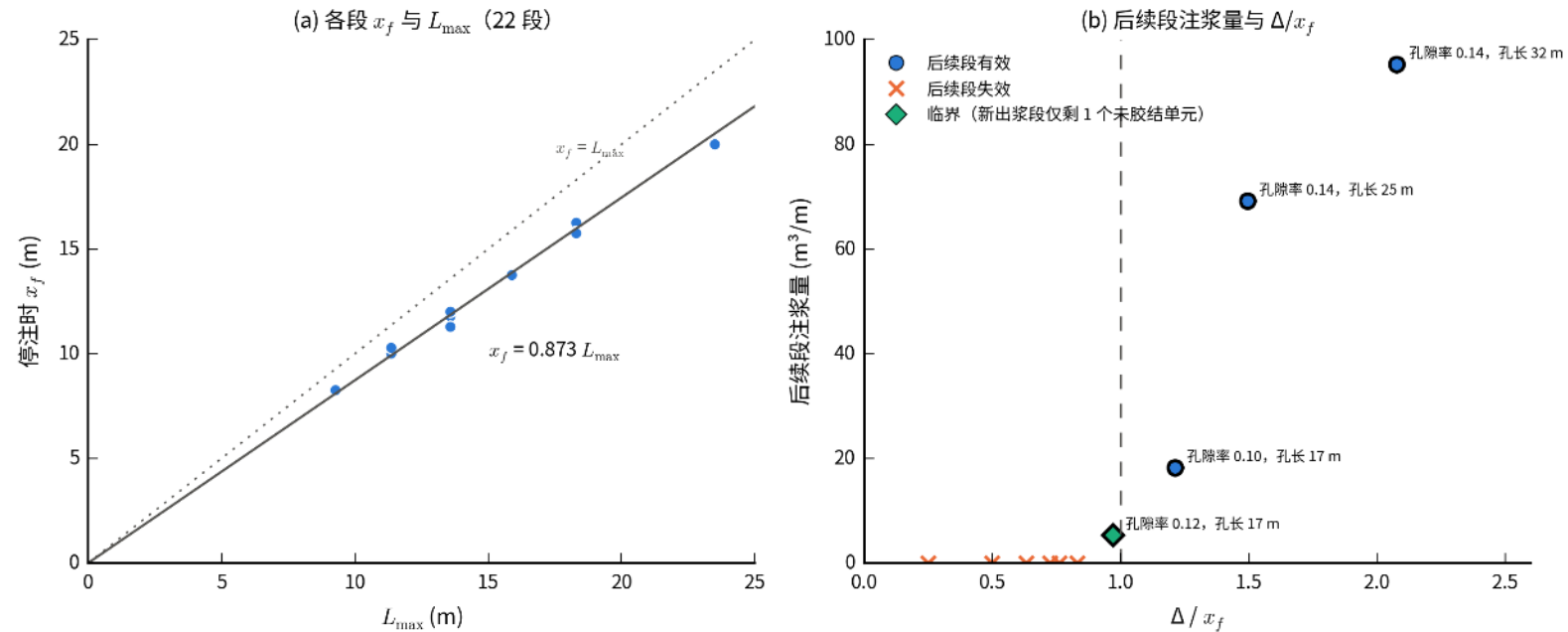


图 4.6-3

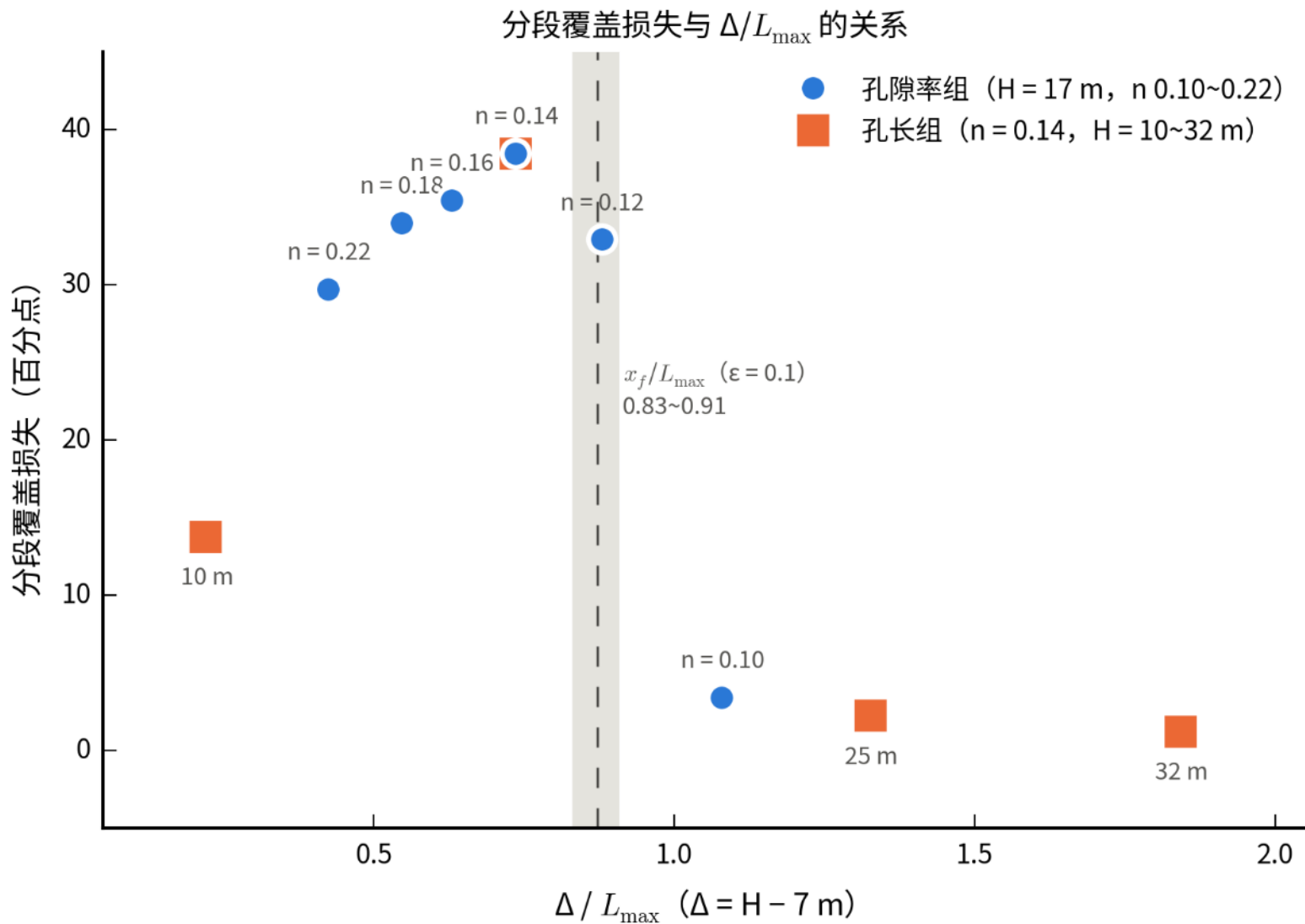


图 4.6-4

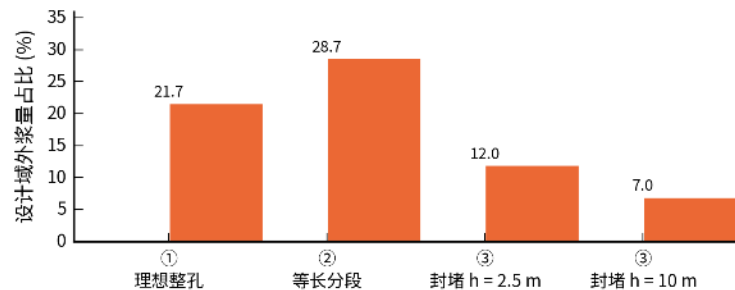
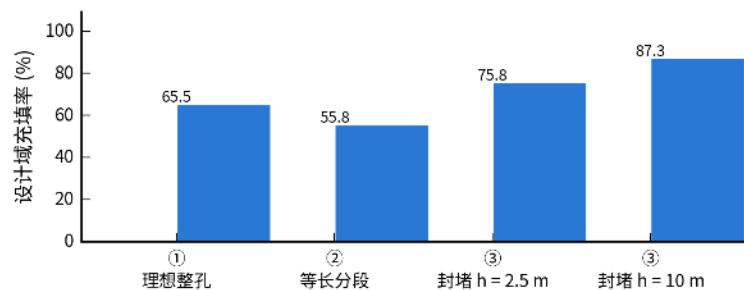
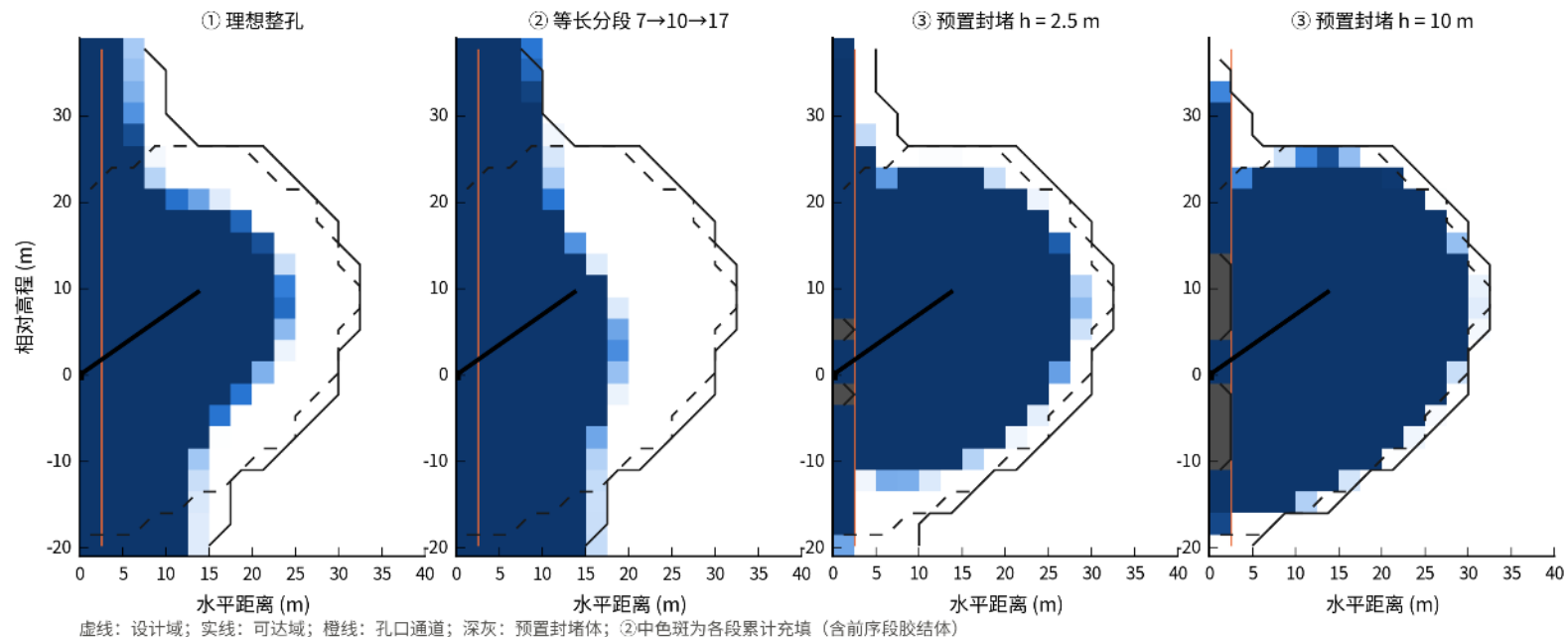
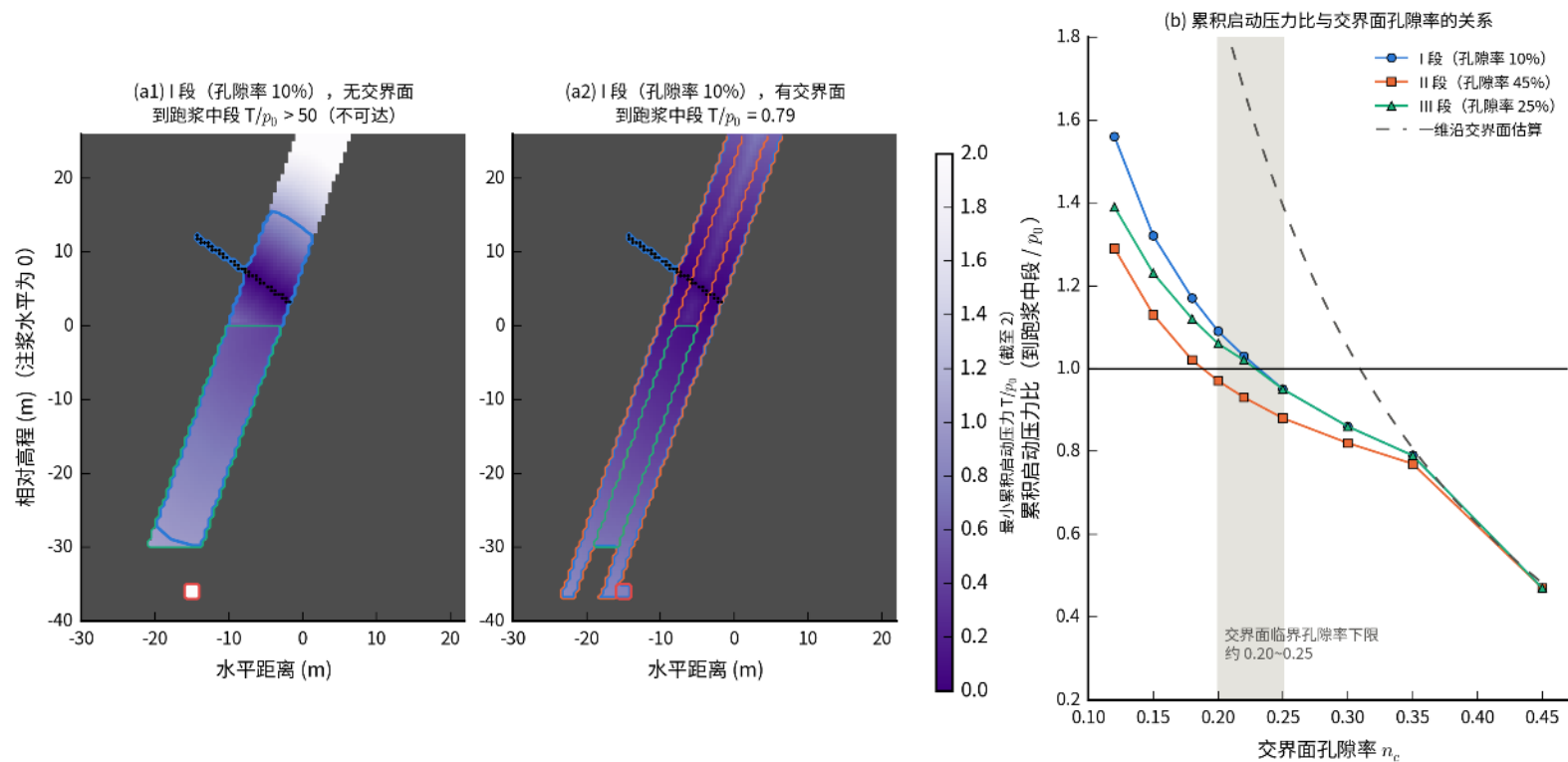
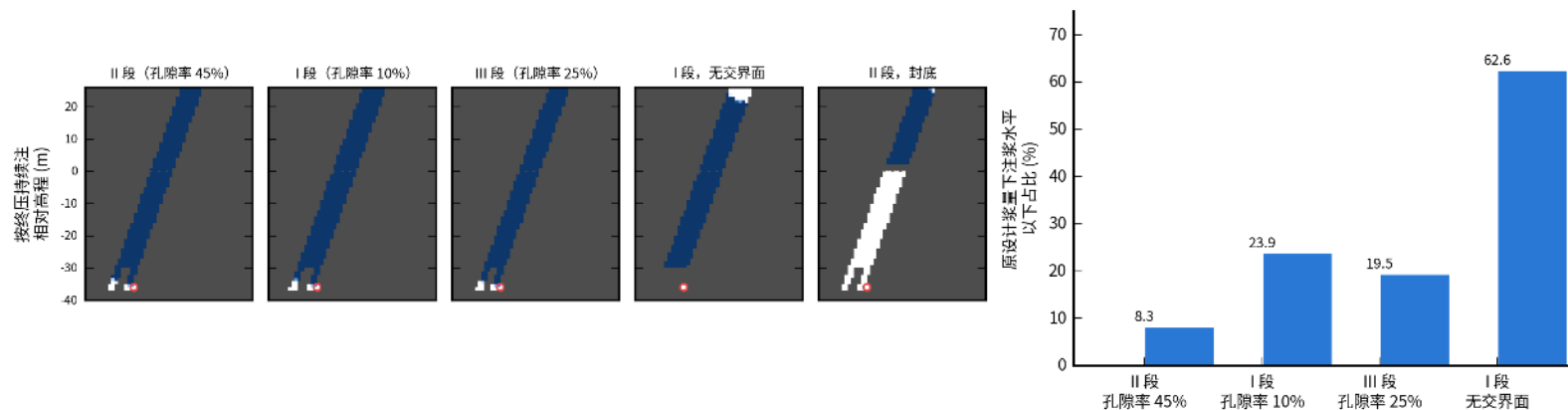
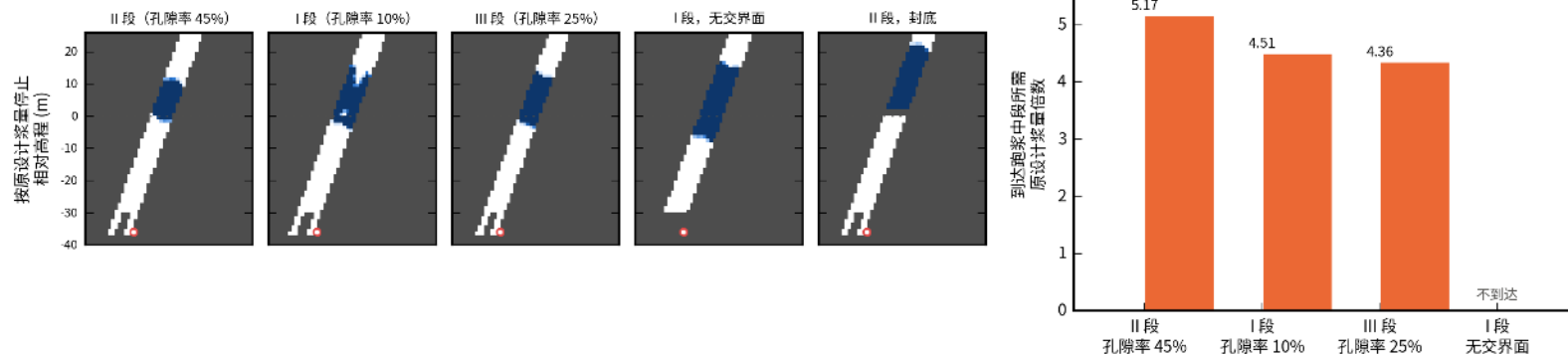


图 4.7-2



蓝线: 可达域边界 ($T = p_0$) ; 橙线: 交界面; 绿线: 存窿矿石; 红框: 跑浆中段; 黑点: 钻孔出浆段; 深灰: 围岩与未采矿体 (不透浆)

图 4.7-3



色斑: 充填域; 深灰: 围岩、未采矿体与封底; 红框: 跑浆中段; 点线: 注浆水平。上排 E 为封底后按变更设计浆量停止。